

POLARIZATION

1. INTRODUCTION

- explain what a WDN is and why it is important
- explain inverse problems and measurement design optimization (MSO)
- explain why the modeled scenario is interesting, and what previous work is lacking
- explain main contribution
- give small numerical study, rate chebyshev point vs GNN solution
- give overview of paper structure

2. DYNAMIC MULTI-START (DMS) FOR ROBUST RADIUS ESTIMATES

2.1. Goal.

For a fixed network and fixed measurement sites, the solver may return very small local optima for W_d although larger values exist. DMS replaces a single run by an adaptive sequence of starts and stores a certificate interval in each artifact.

2.2. Parameters.

- k (consistency): positive integer, default $k = 3$.
- δ (deviation): real number in $(0, 1]$, default $\delta = 0.95$.

2.3. Inputs and outputs.

- Input state: current reference radius r from local search state (not stored in artifact).
- Single-start outputs: radii $r_1, r_2, \dots$ from repeated starts for the same configuration.
- Artifact certificate: two fields
 - lower_bound
 - upper_bound

2.4. One DMS run on a fixed configuration.

Assume one DMS invocation produces single-start radii $r_1, r_2, \dots$. Sort descending as

$$r_1 \geq r_2 \geq r_3 \geq \dots$$

.

DMS can produce two certificate types:

1. No-improvement certificate

If at least one start satisfies $r_i \geq r$, then write

$$\text{lower_bound} := r_i, \quad \text{upper_bound} := -\infty.$$

This certifies that the current configuration is not better than the current reference r (for minimization). Hence this configuration can be discarded safely.

2. Improvement certificate

If at least k starts were made and

$$r_k \geq \delta \cdot r_1,$$

then write

$$\text{lower_bound} := r_1, \quad \text{upper_bound} := r_1.$$

This marks a stable value under the DMS consistency rule and is treated as a relatively confident improvement estimate.

2.5. Cache interaction with current radius r .

When a DMS artifact with $(\text{lower_bound}, \text{upper_bound})$ is considered for caching together with current radius r , cache it only if

$$r < \text{lower_bound} \quad \text{or} \quad r > \text{upper_bound}.$$

Otherwise, run DMS again with the same current r . If a new artifact is produced, update bounds by merging old and new certificates:

$$\begin{aligned} \text{lower_bound} &:= \min(\text{lower_bound_old}, \text{lower_bound_new}), \\ \text{upper_bound} &:= \max(\text{upper_bound_old}, \text{upper_bound_new}). \end{aligned}$$

2.6. Local-search usage.

For each measurement configuration:

- Start with $r = \infty$.
- Call DMS with this r .
- Use returned certificate and update current best radius as local search proceeds.
- For every new candidate configuration, call DMS with the current best r .

This gives an adaptive budget: configurations near the best known radius are re-tested more often, while clearly bad or clearly better candidates are decided faster.

2.7. Practical interpretation.

- Single-start feasibility alone is not treated as fully reliable quality information.
- DMS promotes values that are reproducible across starts.
- The certificate interval separates
 - no-improvement certificates (configuration is worse and can be discarded), and
 - improvement certificates (configuration is likely better and stable under k, δ).

3. IMPLEMENTED W_D MODEL (CODE-TO-MATH)

3.1. Scope.

This chapter documents the currently implemented W_d pipeline in the codebase. We focus on the a-priori W_d branch (not $W_{d(M)}$), which is the default in the solver tab.

3.2. Execution flow for a-priori W_d .

For a fixed site set M :

- The GUI payload sets $\text{MODE} = \text{W_d}$ and force-overrides measurement source to base (no custom measurement instance is used in a-priori W_d).

- The backend builds measurement metadata from base heads and base total demand, but the optimization is solved in equivalence-class mode.
- The solver chooses the xd backend when METHOD = xd.
- Multi-start or Dynamic Multi-Start (DMS) runs several starts and keeps a selected run.
- The reported radius is

$$W_{d(M)} = \|d^A - d^B\|$$

with the configured norm.

3.3. Decision variables (xd backend).

For each pipe e and junction j :

- q_e^A, q_e^B : flows in scenarios A, B .
- x_e^A, x_e^B : headloss variables with nonlinear coupling to flows.
- d_j^A, d_j^B : implied nodal demands from flow balance.

No explicit nodal head variables are optimized in xd mode.

3.4. Implemented optimization problem for W_d .

Given network (V, E) , junction set J , measurement sites $M \subseteq V$, resistance r_e , and exponent n :

Maximize demand separation:

$$\max \|d^A - d^B\|$$

subject to (for both scenarios):

1. Pipe law

$$x_e^s = r_e q_e^s |q_e^s|^{n-1}, \quad s \in \{A, B\}, e \in E.$$

2. Cycle consistency (one equation per cycle row c)

$$\sum_{e \in E} \sigma_{c,e} x_e^A = 0,$$

$$\sum_{e \in E} \sigma_{c,e} x_e^B = 0.$$

3. Measurement equivalence (head-equality-only form)

For each measurement node $m \in M$ a path row is built from reservoir to m :

$$\sum_{e \in E} \pi_{m,e} (x_e^A - x_e^B) = 0.$$

This enforces equal head **differences** to the reservoir, hence equal heads at measured nodes between A and B .

4. Junction demand definitions and lower bounds

$$d_j^s = \text{inflow}_j(q^s) - \text{outflow}_j(q^s), \quad s \in \{A, B\},$$

$$d_j^s \geq \ell_j,$$

where

$$\ell_j = \max(\text{DEMAND_LB}, d_j^{\text{ref}}).$$

Exact implementation of inflow/outflow (per junction j):

$$\begin{aligned} \text{inflow}_j(q^s) &= \sum_{e \in E: \text{end}(e)=j} q_e^s, \\ \text{outflow}_j(q^s) &= \sum_{e \in E: \text{start}(e)=j} q_e^s. \end{aligned}$$

Hence each pipe contributes with a sign induced by its stored orientation ($\text{start}(e), \text{end}(e)$).

If a pipe is incident to the reservoir and to a junction j , it still appears in the equation for d_j^s with the corresponding sign. So reservoir-incident edges are part of the demand definition.

5. A-priori total-demand budget (upper bound)

$$\begin{aligned} \sum_{j \in J} d_j^A &\leq D_{\max}, \\ \sum_{j \in J} d_j^B &\leq D_{\max}, \end{aligned}$$

with

$$D_{\max} = \sum_{j \in J} d_j^{\text{ref}} + \text{EXTRA_DEMAND}.$$

6. Pairwise reservoir outflow equality

$$\text{netOut}(q^A) = \text{netOut}(q^B).$$

with

$$\text{netOut}(q^s) = \sum_{e \in E: \text{start}(e)=r} q_e^s - \sum_{e \in E: \text{end}(e)=r} q_e^s,$$

where r is the selected reservoir node.

3.5. Objective realization.

- For finite p : maximize $\sum_j |d_j^A - d_j^B|^p$.
- For $p = \infty$: maximize epigraph t with $t \geq |d_j^A - d_j^B|$ for all j .

The saved radius is then converted back to $\|d^A - d^B\|$.

3.6. DMS layer (selection over starts).

When DMS is enabled, the above optimization is solved repeatedly with perturbed starts. The algorithm then assigns one of:

- no-improvement certificate,
- improvement certificate,
- inconclusive certificate,

based on consistency/deviation thresholds and reference radius r .

This layer does not change the per-start NLP constraints; it changes acceptance and caching logic.

3.7. Reservoir handling audit.

This section answers whether reservoir conditions are correctly represented.

1. Is reservoir outflow implemented?

- Yes, as a side condition between paired scenarios in a-priori W_d :

$$\text{netOut}(q^A) = \text{netOut}(q^B).$$

- The net outflow expression explicitly sums all reservoir-incident edges with orientation signs.
- In a-priori W_d , reservoir outflow is **not** fixed to a measured scalar value by default; only pair equality is enforced.
- Total demand is also controlled separately through the upper bound $D_{\max}$.

1b. Is “where the flow comes from” preserved when flow is unknown?

- Yes, in the implemented equations: the origin information is carried by
 - directed pipe incidence in $d_j = \text{inflow}_j - \text{outflow}_j$,
 - nonlinear pipe laws, and
 - cycle/path constraints.
- So this is not a demand-only reduced model; flow variables are explicit and source-adjacent edges are represented.
- What is **not** enforced in this branch is a fixed absolute reservoir outflow value (unless explicitly requested), only pairwise equality and global total-demand cap.

3.8. Small worked example: where does source-flow information appear?.

Consider one reservoir r and two junctions 1, 2 with two pipes:

- $e_1 : r \rightarrow 1$
- $e_2 : 1 \rightarrow 2$

Assume these directions are exactly the stored pipe orientations. For one scenario s , the implemented demand definitions are:

$$d_1^s = q_{e_1}^s - q_{e_2}^s,$$

$$d_2^s = q_{e_2}^s.$$

Hence

$$q_{e_2}^s = d_2^s,$$

$$q_{e_1}^s = d_1^s + d_2^s.$$

So the flow on the reservoir-incident edge e_1 is directly tied to downstream demands. This is exactly the information “where the flow comes from” in this oriented model.

The implemented reservoir outflow expression is

$$\text{netOut}(q^s) = q_{e_1}^s,$$

because e_1 starts at r and no edge ends at r in this toy network. Therefore

$$\text{netOut}(q^s) = d_1^s + d_2^s.$$

In the two-scenario W_d model, the code enforces

$$\text{netOut}(q^A) = \text{netOut}(q^B),$$

so

$$d_1^A + d_2^A = d_1^B + d_2^B.$$

This shows two things clearly:

- Reservoir-incident edges are part of the model and carry source-flow information.
- The current a-priori branch constrains this information **relatively** (between A and B), not to a fixed absolute measured value.

How do measurement path constraints interact? If node 2 is a measurement site, xd uses one path equation from r to 2:

$$(x_{e_1}^A + x_{e_2}^A) - (x_{e_1}^B + x_{e_2}^B) = 0.$$

Because $x_e = r_e q_e |q_e|^{n-1}$, this constraint still depends on flows on both the reservoir edge and the downstream edge. So reservoir-edge flow is not discarded there either.

What is **not** encoded by that equation alone is an external absolute reference for “how much must leave the reservoir”. That requires either:

- explicit fixed reservoir outflow, or
- an equivalent absolute condition (if available from data).

2. Is “reservoir is a measurement site” implemented?

- In xd mode, measurement equalities are encoded by path equations from reservoir to each site.
- If the reservoir itself is listed as a measurement site, that path is trivial and contributes no extra equation.
- This is mathematically consistent: head equality at the reference node is tautological.

3. Is absolute reservoir head enforced in xd mode?

- Not as a hard optimization constraint in the xd model.
- xd enforces only head **differences** via x -path equations; the absolute head level is a gauge.
- Reservoir head is used for reconstructing/reporting heads after optimization, not for tightening the feasible set in xd.

4. Important structural note

- Junction demand equations are written only for non-reservoir nodes.
- Reservoir balance is represented only through the explicit net-outflow expression above.
- In the current implementation, one reservoir node is selected as reference in this xd branch.

3.9. Conclusion on reservoir treatment.

- The implementation treats reservoir outflow as a paired side condition (plus global demand budget), which is coherent with the intended equivalence-class variant.
- “Reservoir as measurement site” is handled implicitly and is redundant in xd path-equality formulation.
- If one wants absolute-head anchoring at the reservoir as a strict constraint in xd, that would require an additional explicit anchoring mechanism not currently present in the xd NLP itself.

4. MCMC POSTERIORI SCENARIO GENERATION

4.1. Goal.

We fix measurement data

$$M = (M_S, D),$$

where M_S denotes the measured pressures on the sensor set S and D denotes the measured total reservoir demand.

Given a predictor P , we want to evaluate how good the prediction $P(M)$ is relative to the conditional scenario law induced by the Dirichlet demand model under the measurement constraint M .

The target is therefore not a worst-case quantity, but a posterior distribution on feasible scenarios:

$$\pi(h, d \mid M).$$

The practical plan is to sample this distribution by a Metropolis-Hastings chain whose state is a reduced pressure vector.

4.2. State parametrization.

Let V be the junction set and let $S \subseteq V$ be the measurement set. The pressures on S are fixed by M_S , so only the unobserved pressures can vary.

Choose one additional node v among the unobserved nodes, that is, among the nodes of V that are not in S , as an elimination node. For the first implementation, choosing a central node is a reasonable default because it reduces the risk that the eliminated coordinate produces numerically degenerate pressure differences across long thin parts of the network.

Write U for the unobserved node set, namely the nodes of V that are not in S , and decompose the unknown pressures as

$$h_U = (h_v, h_F),$$

where

$$F$$

denotes the free unobserved nodes after removing the chosen elimination node v from U .

The chain state is then the reduced vector

$$z := h_F \in \mathbb{R}^{|F|}.$$

For a given z , the full pressure vector is reconstructed as follows:

- on S , use the fixed measured values M_S ;
- on F , use the proposed values z ;
- at v , recover h_v from the total-demand constraint.

So the full pressure vector has the form

$$h(z) = (M_S, h_{v(z)}, z).$$

4.3. Recovering the missing pressure by Newton iteration.

Let

$$d(h) = (d_{j(h)})_{j \in V}$$

denote the demand vector reconstructed from the pressure vector h . In the intended posterior pipeline, this reconstruction is assumed to be uniquely determined once h is known.

Define the scalar constraint function

$$g(h_v; z) := \sum_{j \in V} d_{j(M_S, h_v, z)} - D.$$

We recover the missing pressure h_v by solving

$$g(h_v; z) = 0.$$

In practice:

1. start from an initial guess h_v^0 , preferably taken from the predictor state $P(M)$;
2. run a few Newton iterations

$$h_v^{k+1} = h_v^k - \frac{g(h_v^k; z)}{\partial_{h_v} g(h_v^k; z)};$$

3. accept the reconstructed value if the residual is sufficiently small and the resulting demands satisfy the feasibility conditions.

The feasibility filter is:

- $d_{j(h(z))} \geq d_j^0$ for all junctions j ;
- any additional hydraulic admissibility conditions required by the model.

If this fails, the proposal is rejected immediately.

4.4. Posterior density in reduced coordinates.

The original Dirichlet model is a distribution on demand scenarios. Hence the posterior density in pressure coordinates must include the change-of-variables factor from pressures to demands.

Let

$$F(h) := d(h)$$

be the pressure-to-demand map. The full Jacobian is

$$J(h) := \frac{\partial F}{\partial h}.$$

Because measured pressures are fixed and one extra coordinate h_v is eliminated through the scalar constraint, the actual chain runs in the reduced coordinates $z = h_F$. The induced demand map is

$$F_{\text{red}}(z) := F(M_S, h_{v(z)}, z).$$

Its Jacobian is the reduced Jacobian

$$J_{\text{red}}(z) := \frac{\partial F_{\text{red}}}{\partial z}.$$

The posterior density in reduced coordinates is then, up to normalization,

$$\pi_{\text{red}}(z \mid M) = C(M) \pi_{\text{Dir}}(F_{\text{red}}(z)) \times \sqrt{\det(J_{\text{red}}(z)^T J_{\text{red}}(z))} \times 1_{\text{feasible}}(z).$$

Here:

- π_{Dir} is the Dirichlet density expressed on the demand vector;
- $\sqrt{\det(J_{\text{red}}^T J_{\text{red}})}$ is the Gram-determinant factor for the immersion from reduced pressure coordinates into demand space;
- 1_{feasible} is the hard feasibility indicator.

4.5. Exact reduced-Jacobian formula.

Write the full Jacobian columns with respect to the free coordinates and the eliminated coordinate as

$$J(h) = \begin{bmatrix} J_{F(h)} & J_{v(h)} \end{bmatrix},$$

where J_F contains the columns corresponding to $z = h_F$ and J_v is the column corresponding to h_v .

Differentiate the scalar constraint

$$g(h_{v(z)}; z) = 0$$

with respect to z . This gives

$$\partial_z g + \partial_{h_v} g \partial_z h_v = 0,$$

so

$$\partial_z h_v = -\frac{\partial_z g}{\partial_{h_v} g}.$$

By the chain rule,

$$J_{\text{red}} = J_F + J_v(\partial_z h_v).$$

Substituting the previous expression yields

$$J_{\text{red}} = J_F - J_v \frac{\partial_z g}{\partial_{h_v} g}.$$

This is the precise reduced-coordinate formula. Relative to the free-column Jacobian J_F , the correction term is of rank one, because it is an outer product of the column vector J_v with the row vector

$$\frac{\partial_z g}{\partial_{h_v} g}.$$

Passing from the full Jacobian to the reduced Jacobian is a rank-one modification once the eliminated pressure is recovered through the total-demand constraint.

4.6. Removing one demand row.

Often one does not want to use the full rectangular Jacobian from reduced pressure coordinates into all demands, because the total-demand constraint introduces one scalar dependence between the demand components. A convenient square version is obtained by deleting the row corresponding to the chosen node v .

Let

$$F^{-v}(h) := \left(d_{j(h)}\right)_{j \neq v},$$

and let

$$J_{\text{red},-v}$$

be the corresponding reduced Jacobian. Then the same derivation gives

$$J_{\text{red},-v} = J_F^{-v} - J_v^{-v} \frac{\partial_z g}{\partial_{h_v} g}.$$

Again, this is a rank-one modification of the Jacobian obtained after removing the row of the chosen node and keeping only the free pressure coordinates.

This square reduced Jacobian is often the cleanest object for local volume factors, and its Gram determinant is

$$\det\left((J_{\text{red},-v})^T J_{\text{red},-v}\right).$$

4.7. Metropolis-Hastings chain.

With the reduced state z , one MCMC step has the following form.

1. Current state: z_k .
2. Proposal: sample

$$z' = z_k + \eta,$$

where η is centered Gaussian noise.

3. Reconstruct $h_v(z')$ by Newton iteration.
4. Reconstruct the full pressure vector $h(z')$ and then the demand vector $d(z')$.
5. If feasibility fails, reject.
6. Otherwise compute the log target density

$$\log \pi_{\text{red}}(z' \mid M)$$

using the Dirichlet term and the reduced Gram determinant.

7. Accept with the usual Metropolis-Hastings probability.

For a symmetric Gaussian proposal, the acceptance ratio is simply

$$\alpha(z, z') = \min\left(1, \frac{\pi_{\text{red}}(z' | M)}{\pi_{\text{red}}(z | M)}\right).$$

4.8. Initialization and burn-in.

The natural initialization is the predictor output $P(M)$.

More precisely:

- use the predicted pressure vector to initialize the free coordinates z ;
- use its value at the elimination node as the initial Newton guess for recovering h_v ;
- start the chain there and discard an initial burn-in segment.

After burn-in, the retained samples provide posterior scenarios compatible with the measurement instance M .

4.9. Performance tricks (practical and standard).

The construction above is standard Metropolis-Hastings in transformed coordinates [1, 2]. The following implementation tricks are typically decisive for robustness and speed.

1. Evaluate the target in log form. Instead of evaluating $\pi_{\text{red}}(z | M)$ directly, evaluate

$$\ell(z) := \log \pi_{\text{red}}(z | M)$$

and use

$$\log \alpha(z, z') = \min(0, \ell(z') - \ell(z)).$$

This avoids underflow/overflow and turns products into sums, which is numerically much more stable [3, 4].

2. Use determinant computations through factorization, not through raw `det`. For square reduced Jacobian, compute $\log |\det J_{\text{red},v}|$ via an LU factorization with pivoting. For Gram forms, compute $\log \det((J^T J))$ via Cholesky of $(J^T J)$ if positive definite. These are standard stable linear algebra pathways [5, 6].
3. Exploit the rank-one structure. Since

$$J_{\text{red},v} = A - ur^T,$$

one may use the matrix determinant lemma

$$\det(A - ur^T) = \det(A)(1 - r^T A^{-1}u)$$

and evaluate the correction with one linear solve after reusing a factorization of A . This can reduce cost substantially when many proposals share the same structural sparsity pattern [5, 7].

4. Reuse local linearization artifacts across proposals. In each MH step, the expensive operations are typically Newton recovery of h_v and Jacobian-

related solves. Reusing previous iterate values as warm starts (for Newton and linear solves) is usually a large constant-factor gain.

5. Tune proposal scale to avoid random-walk degeneracy. For random-walk Metropolis in moderate/high dimension, too small variance mixes slowly and too large variance causes frequent rejection. A common target regime is the classical random-walk acceptance window around the 0.2 to 0.3 range (with the 0.234 asymptotic heuristic) [8].
6. Monitor effective sampling, not only raw sample count. Report acceptance rate, autocorrelation diagnostics, and effective sample size (ESS). This is standard MCMC practice and is usually more informative than the number of retained states alone [4, 9].

4.10. Why this is the right first formulation.

This construction separates the problem into three layers:

- a geometric layer: pressures are the sampling coordinates;
- a physical layer: demands are reconstructed from pressures and filtered by feasibility;
- a probabilistic layer: the Dirichlet prior on demands is pulled back to the reduced pressure space.

This is the correct starting point for posterior predictor evaluation, because it produces samples from the conditional scenario law instead of only an extremal scenario.

Once this posterior sampler is available, one can compare $P(M)$ to posterior summaries such as:

- posterior mean,
- posterior quantiles,
- posterior credible bands,
- posterior log-density at $P(M)$.

5. SAMPLING THE POSTERIOR: MCMC, ABC, AND EXACT ENUMERATION

The previous chapter constructed a Metropolis-Hastings sampler for the posterior $\pi(h, d \mid M)$. This chapter reports what happens when that sampler is applied to networks larger than the small benchmarks, why it fails there, and which two cheaper methods replace it. The conclusion is that the choice of method is governed by a single scalar — the ratio of the prior-predictive spread of the sensor readings to the assumed sensor noise — and that on the networks of interest that ratio almost always favours something other than MCMC.

5.1. Setting.

Throughout, $\mathcal{J}$ is the junction set, $\mathcal{Y} \subseteq \mathcal{J}$ the sensor set, and an observation is $z = (D^z, h_y^z)$. The sensor likelihood is Gaussian of width ε ,

$$P(z \mid s) \propto \exp\left(-\frac{1}{2\varepsilon^2} \|h_y^s - h_y^z\|^2\right),$$

and the prior P is one of: a Dirichlet split of an extra demand Δ over all junctions, a Gaussian $N(d^0, \text{diag}(\sigma^2))$ around the base demands, or the **single-leak** prior

$$v \sim \text{Unif}(\mathcal{J}), \quad \lambda \sim \text{Unif}([0, \Delta]), \quad d = d^0 + \lambda e_v.$$

Definition: The **signal-to-noise ratio** of a sensor $y \in \mathcal{Y}$ under a prior P is

$$\text{SNR}_y := \frac{\text{std}_{s \sim P}[h_y^s]}{\varepsilon},$$

the spread of that sensor’s reading across the prior, measured in units of the sensor noise.

The whole of this chapter can be read as a study of this quantity. When $\text{SNR} \lesssim 1$ the likelihood is flatter than the prior and the measurement is weakly informative; when $\text{SNR} \gtrsim 1$ the posterior concentrates into a thin sliver of the prior support.

5.2. Why MCMC does not work on larger networks.

5.2.1. A representative failure.

Take L-TOWN_C (BattLeDIM Area C): 92 junctions, 109 pipes, one reservoir, three pressure sensors. Sampling the Gaussian-prior posterior with the affine-invariant ensemble sampler, 2000 burn-in and 2500 retained iterations with $2 \cdot 91 + 2 = 184$ walkers, gives

wall clock	4568 s
minimum ESS	1010
ESS per second	0.2
maximum split- $\hat{R}$	2.96
mean-disagreement	0.90σ

A split- $\hat{R}$ of 2.96 after 76 minutes is not a chain that needs a little more time; it is a chain whose walkers have not mixed at all. Both diagnostics that were introduced precisely to catch this — $\hat{R}$ and the per-chain mean disagreement — fire.

5.2.2. The failure is structural, not a tuning problem.

It is tempting to read this as insufficient tuning. It is not. Linearising the forward map at the base state gives the Jacobian $J = \partial h_y / \partial d \in \mathbb{R}^{3 \times 92}$, whose singular values are 453, 102, 53 and nothing else: J has rank 3. Conditioning a Gaussian prior on three linear functionals leaves the remaining 89 directions **exactly** at their prior. The exact linear-Gaussian posterior covariance

$$\Sigma_{\text{post}} = (\Sigma^{-1} + J^\top J / \varepsilon^2)^{-1}$$

gives a total information content, summed over junctions, of 2.02 out of a possible 92 even in the noiseless limit $\varepsilon \rightarrow 0$, and 0.085 at the realistic $\varepsilon = 0.015$ m.

So the posterior is, in 89 of 92 directions, identical to the prior. A sampler asked to explore such a distribution in 91 dimensions is doing an enormous amount of work to reproduce something already known in closed form, and the thin remainder is what it must resolve. That is a bad trade.

5.2.3. *The dangerous part: a non-converged chain gives a plausible wrong answer.*

The failure would be benign if it were loud. It is not. The unconverged chain reported a per-junction variance reduction of median 0.017 with a maximum of 0.77 — that is, it claimed one junction’s demand had been pinned down to 23% of its prior uncertainty. The exact linear-Gaussian computation gives median 0.0001 and maximum 0.046, and **no** junction above 10%.

Observation 5.2.3.1: A stuck ensemble under-disperses: its walkers have not explored the prior, so the empirical spread of the samples is too small, and the inferred variance reduction is correspondingly too large. Non-convergence therefore biases the reported uncertainty in the **optimistic** direction. A posterior width taken from an unconverged chain will overstate how much the sensors have taught us.

This is the strongest argument against relying on MCMC here: the failure mode produces a confident, wrong, and superficially reasonable answer.

5.3. Approximate Bayesian Computation.

5.3.1. *The method.*

Draw $s_1, \dots, s_N$ from the prior, forward-solve each, and weight by the likelihood,

$$w_i \propto \exp\left(-\frac{1}{2\varepsilon^2} \|h_y^{s_i} - h_y^z\|^2\right), \quad \sum_i w_i = 1.$$

Posterior expectations are weighted sums, and the quality of the approximation is measured by the effective sample size $\text{ESS} = 1 / \sum_i w_i^2$.

The textbook objection is the curse of dimensionality: acceptance scales like $(1/\varepsilon)^{|y|}$. The point of the SNR is that this curse is **conditional**. If ε exceeds the spread of the readings across the prior, almost every draw is consistent with the observation and the weights are nearly uniform.

5.3.2. *It is essentially free in the weakly-informative regime.*

On L-TOWN_C the prior-predictive spreads are 0.0034, 0.0051 and 0.0025 m against $\varepsilon = 0.015$ m, so $\text{SNR} \in [0.17, 0.34]$. Running ABC on the identical model that the MCMC above sampled:

	ABC	MCMC
draws / iterations	8000	4500
wall clock	147 s	4568 s

	ABC	MCMC
ESS	7968	1010
ESS / N	0.996	—
ESS per second	54.3	0.2
converged	yes	no ($\hat{R} = 2.96$)

ABC is $270 \times$ more efficient per second, and it **converges**: an ESS of 7968 from 8000 draws means the samples are essentially independent. It also reproduces the exact linear-Gaussian answer, where the MCMC did not.

5.3.3. Where ABC does break down.

The curse is real when the measurement is informative. On Kadu (24 junctions, 4 sensors, prior-predictive spread 0.426 m) the realised efficiency across ε is

ε (m)	SNR	ESS (of 3000)	ESS / N
0.5	0.85	833	0.28
0.2	2.13	147	0.049
0.1	4.26	44	0.015
0.05	8.52	9.5	0.003
0.02	21.3	1.1	0.0004

Below $\varepsilon \approx 0.05$ m the weights collapse onto a single draw. Two remarks make this less damaging than it appears.

First, **MCMC fails in the same place**: on Kadu with four sensors the ensemble sampler reaches $\hat{R} \approx 3.4$ at $\varepsilon = 0.05$ and mixes acceptably only at $\varepsilon \approx 0.5$. The wall at small ε is a property of the posterior, not of the algorithm; it is the statement that the measurement-consistent demand set is genuinely tiny.

Second, the small- ε regime is not the physically relevant one. Kadu's sensors sit at ≈ 98 m of head; a pressure transducer at 0.1–0.5% of a 100 m full scale contributes 0.1–0.5 m, and the nominal-versus-real model mismatch of $\pm 10\%$ in demands and pipe parameters contributes a further median 0.11 m and 95th percentile 0.45 m. A defensible budget is $\varepsilon \in [0.2, 0.5]$ m, which is exactly the regime where ABC is usable.

Observation 5.3.3.1: The honest reading of the table is uncomfortable and worth stating plainly: ABC is efficient at $\varepsilon = 0.5$ m **because** the measurement is barely informative there. Kadu's entire network head spread is 2.00 m, so $\varepsilon = 0.5$ m is a quarter of the whole signal. Efficiency of the sampler and informativeness of the data trade off against one another; there is no configuration in which both are large.

5.3.4. ABC needs $\varepsilon > 0$.

For a prior with a density, the set of demand vectors reproducing the sensor readings exactly has measure zero, so at $\varepsilon = 0$ the acceptance probability is exactly zero. ABC therefore **requires** a positive noise level. This is not a defect of the implementation; it is why the Dirac likelihood is listed as inapplicable to ABC. It is also precisely the gap that the next section fills.

5.4. The exact solver.

5.4.1. The single-leak prior collapses.

Under the single-leak prior the parameter space is not high-dimensional: it is one discrete label v and one scalar λ . Geometrically the support is a union of $|\mathcal{J}|$ line segments radiating from d^0 , a one-dimensional set with $|\mathcal{J}|$ branches, and one can simply walk it. Two variants arise.

Algorithm 5.4.1.1: Variant A — total demand observed. Since $d = d^0 + \lambda e_v$ implies $\mathbb{1}^\top d = D^0 + \lambda$, the observed total demand determines $\lambda^* = D^z - D^0$ outright. Then for each candidate v :

1. forward-solve $h_y(d^0 + \lambda^* e_v)$;
2. form the residual $r_v = \|h_y(v) - h_y^z\|$;
3. set $P(v \mid z) \propto K_\varepsilon(r_v)$ and normalise over the $|\mathcal{J}|$ candidates.

Cost: $|\mathcal{J}|$ forward solves.

Algorithm 5.4.1.2: Variant B — total demand unknown. Now λ must be inferred. For each candidate v :

1. solve $h_y(d^0 + \lambda e_v) = h_y^z$ for λ_v (the sensor head is monotone in λ);
2. form r_v as above;
3. set

$$P(v \mid z) \propto p(\lambda_v) \cdot \|\partial h_y / \partial \lambda\|^{-1} \cdot K_\varepsilon(r_v)$$

and normalise.

The Jacobian factor in Variant B is essential and easy to miss. Conditioning on $h_y(v, \lambda) = h_y^z$ pins λ , and the induced density on the discrete label v picks up the change-of-variables term $\|\partial h_y / \partial \lambda\|^{-1}$: a candidate whose sensor reading responds **slowly** to λ explains a whole neighbourhood of observations, and so deserves more posterior mass than one that responds sharply. Since $p(\lambda_v)$ is constant under the uniform prior and $K_\varepsilon(r_v)$ is identical across all exactly-consistent candidates, this factor is the **only** thing that makes the posterior non-uniform. Omitting it silently returns a uniform distribution over the consistent candidates. It is the same

Gram-determinant correction that appears in the pressure-space and exact-demand MCMC formulations.

5.4.2. *Cost.*

Variant A costs $|\mathcal{J}|$ forward solves. Variant B costs $|\mathcal{J}|$ scalar root-solves; with a secant iteration these converge in about 6 solves each, against 62 for a 60-step bisection, at a cost in λ of $\approx 4 \cdot 10^{-6}$ — far below the $\approx 3 \cdot 10^{-5}$ m precision to which the hydraulic solver reports heads. On a 24-junction network one full posterior is a matter of seconds.

5.4.3. *When it is justified.*

1. **It is exact.** There is no Monte-Carlo error, no burn-in, no proposal to tune, and no convergence diagnostic to interpret. The result is reproducible to solver precision.
2. **It works at $\varepsilon = 0$.** With a finite candidate set there is no measure-zero obstruction, so the noiseless case — inaccessible to ABC — is available. On L-TOWN_C with exact readings this yields 98% exact leak localisation even for the smallest BattLeDIM leak, against 13% with readings rounded to the challenge’s two decimal places.
3. **It is an oracle.** Because it is exact, it is a strictly better reference for validating samplers than an ABC oracle, which carries its own noise.
4. **It makes sensor placement affordable.** Optimising over sensor sets requires one posterior per candidate set. At seconds per posterior that search is tractable; at one MCMC run per posterior it is not.

5.4.4. *When it is not justified.*

The method buys its exactness by assuming a very restrictive prior, and the assumption should be stated rather than hidden.

1. **It requires structure.** The prior must be discrete or have very few continuous parameters. A free demand field over all junctions returns us to the continuous case.
2. **It does not scale in the number of leaks.** For k simultaneous leaks one enumerates $\binom{|\mathcal{J}|}{k}$ node sets. For $|\mathcal{J}| = 92$, $k = 2$ is 4186 pairs and still feasible; $k = 3$ is 125{,}000 triples and the approach loses.
3. **The prior is an assumption about the world.** A single leak of bounded magnitude at a node is a modelling choice. The posterior is exact **given** that choice, and the exactness must not be mistaken for freedom from modelling error.
4. **Real leaks are on pipes, not nodes,** and the nominal model differs from the real network. In particular, on L-TOWN_C the 98% localisation above degrades to 26% once the observation is generated from a network perturbed by the $\pm 10\%$ the challenge specifies — so the exactness of the inference is not the binding constraint on accuracy; model calibration is.

5.5. Perturbing the base demands: a hierarchical prior.

Both priors used so far treat the base demands d^0 as known. They are not: the nominal model differs from the real network, and on the BattLeDIM benchmark

the base demands are randomised by $\pm 10\%$. A more honest prior perturbs them and then places a leak on top,

$$\tilde{d}^0 \sim N(d^0, \Sigma_0), \quad v \sim \text{Unif}(\mathcal{J}), \quad \lambda \sim \text{Unif}([0, \Delta]), \quad d = \tilde{d}^0 + \lambda e_v.$$

This is attractive because the enumeration of the previous section still applies **conditionally**: for a fixed $\tilde{d}^0$ the candidate set is again finite and λ is again recovered from the sensor constraint. The question is how to combine the runs.

5.5.1. *The correct estimator sums evidence, not posteriors.*

Marginalising the nuisance parameter,

$$P(v \mid z) \propto P(v) \int P(z \mid v, \tilde{d}^0) P(\tilde{d}^0) d\tilde{d}^0.$$

For a fixed draw $\tilde{d}_m^0$, the inner λ -integral is exactly the quantity the enumeration already computes. Write that per-draw **evidence** for candidate v as

$$w_{m,v} := p(\lambda_{m,v}) \cdot \|\partial h_y / \partial \lambda\|^{-1} \cdot K_\varepsilon(r_{m,v}),$$

with $\lambda_{m,v}$ and $r_{m,v}$ the solved magnitude and residual for candidate v under base $\tilde{d}_m^0$. Then the Monte-Carlo estimator of the posterior is

Algorithm 5.5.1.1: Rao-Blackwellised enumeration.

1. draw $\tilde{d}_1^0, \dots, \tilde{d}_M^0 \sim N(d^0, \Sigma_0)$;
2. for each m , run the enumeration and record the **unnormalised** $w_{m,v}$;
3. set

$$P(v \mid z) \propto \sum_{m=1}^M w_{m,v}$$

and normalise once, at the end.

The tempting alternative is to normalise each run separately and average the resulting posteriors,

$$P_{\text{naive}}(v \mid z) := \frac{1}{M} \sum_{m=1}^M \frac{w_{m,v}}{\sum_{u \in \mathcal{J}} w_{m,u}}.$$

This is wrong. Writing $E_m := \sum_u w_{m,u}$ for the total evidence of draw m , the correct estimator is

$$P(v \mid z) \propto \sum_m E_m \cdot P(v \mid z, \tilde{d}_m^0),$$

whereas the naive one is $M^{-1} \sum_m P(v \mid z, \tilde{d}_m^0)$.

Observation 5.5.1.2: The naive estimator is the correct one with the evidence weights E_m replaced by uniform weights. It is therefore exact if and only if every perturbed base explains the observation equally well. That is precisely what $\Sigma_0 > 0$ destroys: some draws of $\tilde{d}^0$ fit the sensor readings far better than others, and normalising per draw discards exactly that information. The discrepancy grows with Σ_0 , i.e. with the very model uncertainty the construction was introduced to represent.

5.5.2. *Why this is the collapsed Gibbs structure.*

The state is the pair $(v, \tilde{d}^0)$: one discrete label and one continuous nuisance vector. An ordinary Gibbs sampler would alternate

$$v^{(t+1)} \sim P(v \mid \tilde{d}^{0,(t)}, z), \quad \tilde{d}^{0,(t+1)} \sim P(\tilde{d}^0 \mid v^{(t+1)}, z).$$

Collapsing — equivalently, Rao-Blackwellising — means replacing the sampling of a component by its analytic conditional wherever that conditional is available in closed form. Here it is: v ranges over the finite set $\mathcal{J}$ and $P(v \mid \tilde{d}^0, z)$ is exactly what one enumeration pass returns. So v never has to be sampled at all. For any statistic f ,

$$\mathbb{E}[f(v) \mid z] = \mathbb{E}_{\tilde{d}^0 \mid z}[\mathbb{E}[f(v) \mid \tilde{d}^0, z]],$$

and the inner expectation is evaluated exactly rather than estimated. By the Rao-Blackwell theorem, replacing a sampled indicator by its conditional expectation cannot increase the variance of the estimator, so this dominates any scheme that draws v at random — including plain ABC over the joint prior, which must explore the discrete dimension by chance.

Two honest qualifications on the terminology. First, the scheme above draws $\tilde{d}^0$ from its **prior** and reweights by evidence; that is Rao-Blackwellised importance sampling, not a Gibbs sampler, because no Markov chain is involved. It becomes collapsed Gibbs proper when the prior draws stop being efficient — large Σ_0 or small ε — and one instead samples $\tilde{d}^0 \sim P(\tilde{d}^0 \mid z)$ by MCMC while continuing to enumerate v exactly at every step. The discrete component is collapsed in both cases; only the machinery for the continuous component changes. Second, the argument for exactness rests on the change-of-variables factor of the previous section: without it each $w_{m,v}$ is wrong by a v -dependent constant, and summing wrong weights does not produce a right answer.

5.5.3. *Status.*

The construction is derived here but not yet measured. The experiment that settles it compares, on identical observations whose truth is generated from a **perturbed** base so that the model mismatch is real: (i) the naive average of normalised posteriors, (ii) the evidence-weighted sum above, and (iii) ABC over the full hierarchical prior as an unbiased reference. That would quantify both how far the naive estimator is off and whether the Rao-Blackwellised estimator beats plain

ABC per forward solve, which the Rao-Blackwell theorem predicts but does not quantify.

5.6. Cross-validation of the two methods.

ABC and enumeration are, structurally, the same quadrature applied to

$$\text{posterior weight}_k \propto \text{prior weight}_k \cdot K_\varepsilon(\text{residual}_k),$$

differing only in whether the candidates are deterministic (solutions of the structural constraint) or Monte-Carlo (prior draws). They therefore validate one another: agreement is evidence that both are correct, since they share no failure mode.

On Alperovits, averaged over 30 random observations under the single-leak prior with $\varepsilon = 0.05$ m, the mean per-junction absolute error of the posterior mean is

method	Alperovits, $\varepsilon = 0.05$	Kadu, $\varepsilon = 0.2$
ABC posterior mean	0.1670	0.0027
exact posterior mean	0.1673	0.0006
trained GNN	0.3257	0.0817

On Alperovits, ABC and the exact posterior agree to 0.2% — the mutual validation one hopes for. On Kadu they do **not**: ABC’s mean error is more than four times the exact one. This is not a contradiction but a direct illustration of the earlier efficiency table. With $\varepsilon = 0.2$ m on Kadu the realised ESS is of order tens out of 4000 draws, so the ABC posterior mean is itself a noisy estimate; the enumeration, being exact, is not. The disagreement is a measurement of ABC’s Monte-Carlo error, not evidence against either method, and it is precisely the regime in which the exact solver earns its place.

5.6.1. Reading these numbers: irreducible error versus excess.

All three rows above are scored the same way — each method is collapsed to a single predicted demand vector and compared against the truth, with the posterior **mean** serving as the prediction for the two Bayesian methods. The posterior mean is the Bayes-optimal point estimate under squared loss, so this is the right way to ask how much worse a predictor is than the best one available. But the rows do not mean the same thing.

The exact posterior’s error is not a deficiency of the method. Several demand vectors explain the same sensor reading equally well, so no estimator can be right; the best possible one sits among them and is still wrong by some amount. That amount is the **irreducible** error — a property of the network, the sensor set and the prior, not of any algorithm. A predictor’s total error decomposes as

$$\underbrace{|F(z) - d|}_{\text{total}} \approx \underbrace{|\mathbb{E}[d | z] - d|}_{\text{irreducible}} + \underbrace{|F(z) - \mathbb{E}[d | z]|}_{\text{excess, the predictor's own}},$$

and only the second term is the predictor’s responsibility.

network	irreducible	GNN total	GNN excess
Alperovits	0.1671	0.3257	≈ 0.159
Kadu	0.0006	0.0817	≈ 0.081

Quoting ratios — “twice as bad”, “thirty times as bad” — misleads in both directions. On Alperovits roughly half the GNN’s error is ambiguity it could never have avoided, so the factor of two overstates its shortcoming. On Kadu the data very nearly determines the answer (irreducible error 0.0006), so essentially **all** of the GNN’s error is its own, and the factor of thirty understates the situation rather than exaggerating it.

Observation 5.6.1.1: A second caveat attaches to the GNN row specifically, and it works in the predictor’s favour. The trained model outputs **pressures**; its demands are obtained afterwards by the algebraic inversion $q_e = \text{sign}(\Delta h)(|\Delta h|/r_e)^{1/\kappa}$ followed by nodal mass balance. The two posterior methods never invert pressures — they work in demand coordinates throughout. What is tabulated as “GNN error” is therefore the error of **predictor plus reconstruction**, and the reconstruction is the ill-conditioned direction documented earlier. It is a fair assessment of the deployed pipeline, but it is not an isolated measurement of the learned component, and the excess above should not be attributed to the network without separating the two.

The obvious objection to the comparison is that the model was trained on Dirichlet scenarios and is therefore out of distribution on a single-leak test, so that these factors merely measure distribution shift. Repeating the experiment under the **training** prior $d = d^0 + f\Delta\alpha$, $f \sim \text{Unif}([0, 1])$, $\alpha \sim \text{Dir}(1, \dots, 1)$, refutes it:

network	prior	ABC	GNN
Alperovits	single-leak	0.1670	0.3257
Alperovits	Dirichlet (training law)	0.0700	0.2821
Kadu	single-leak	0.0027	0.0817
Kadu	Dirichlet (training law)	0.0024	0.0734

In distribution the GNN does not improve relative to the posterior; on Alperovits the gap **widens**, from a factor of 1.95 to a factor of 4.0. The reason is visible in the columns: the Dirichlet posterior is markedly tighter than the single-leak one ($0.1670 \rightarrow 0.0700$), while the GNN barely moves ($0.3257 \rightarrow 0.2821$). The measurement carries more information under the training law, and the predictor does not exploit it. The deficiency is therefore a property of the predictor, not an artefact of the test distribution.

Note also that under a continuous prior there is no exact panel: the support is a continuum rather than a finite candidate set, so ABC is the only available reference

and nothing independent validates it. Given the low ESS on Kadu, its Dirichlet figures should be read with the corresponding caution.

One further observation supports the reconstruction caveat above: on Kadu the GNN’s error is concentrated at junction 3, which is the **only reservoir-adjacent junction** in the network. Its reconstructed demand is the difference of the two flows on either side of it, the first of which is the entire network inflow — so a small nodal demand is computed as a small difference of large numbers, and any head error there is amplified by roughly the ratio of inflow to nodal demand. Junction 3 is also a sensor whose prior-predictive reading spread is 0.000 m, i.e. hydraulically insensitive to demand, contributing nothing to the measurement. Neither posterior method shows any such spike, because neither inverts pressures.

5.7. Summary.

regime	character	method
$\text{SNR} \lesssim 1$	likelihood flatter than prior	ABC, nearly free
$\text{SNR} \gtrsim 1$	posterior concentrated	enumeration if the prior has structure
$\varepsilon = 0$	noiseless	enumeration only
high-dim. continuous prior, sharp likelihood	genuinely hard	MCMC, if anything

The band in which MCMC is the right tool — a high-dimensional continuous prior together with a sharp likelihood — is narrow, and it coincides with the band in which every method struggles. On the networks studied here the measurement is either weak enough that ABC is essentially free, or the prior is structured enough that the posterior can be written down.

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